

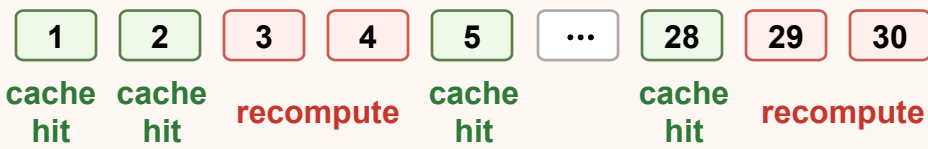
# Challenges

# Proposed Co-Design

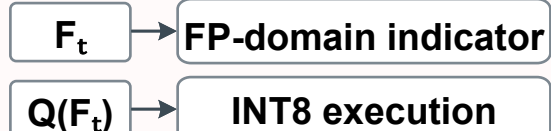
# Benefits / Results

## 1 Decision-Domain Mismatch in Cache+Quant

Illustrative Denoising timeline (30 steps)



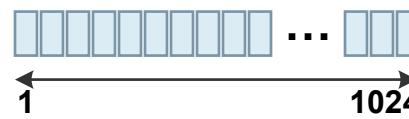
FP indicator  $\neq$  INT8-domain relevance



## 2 Matrix-Axis Mismatch in Factorized Attention

Spatial: 38 (19x2) contexts  $\times$  1024-token rows ( $L=1024$ )

Temporal: 2048 (1024x2) contexts  $\times$  19-token rows ( $L=19$ )



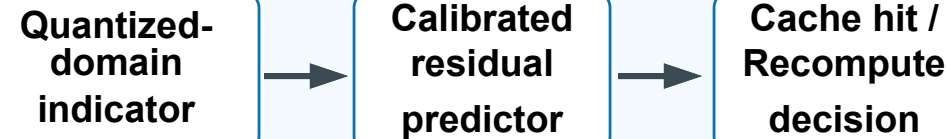
54 $\times$  more contexts

Per head:  $d = 72$ ;  $L = 1024$  (spatial), 19 (temporal)

$QK^T$ :  $[1, d] \times [d, L] \rightarrow L$  on output axis

PV:  $[1, L] \times [L, d] \rightarrow L$  on reduction axis

## 1 QAC: Quantization-Aligned Caching



## 2 DARE: Dimension-Adaptive Engine

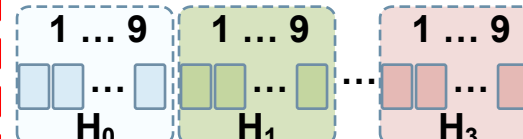
### ① $QK^T$ output-axis Adaptation (e.g., $T=19$ )

Baseline: 1 head  $\times$  36 lanes



19/36=52.8%

Proposed: 4 heads  $\times$  9 lanes



3 tiles/head  $\rightarrow$  70.4%

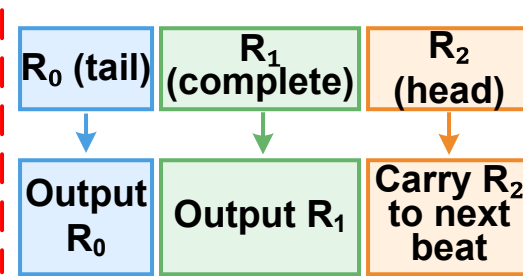
### ② PV reduction-axis Adaptation (e.g., $T=19$ )

Baseline: padding for short rows

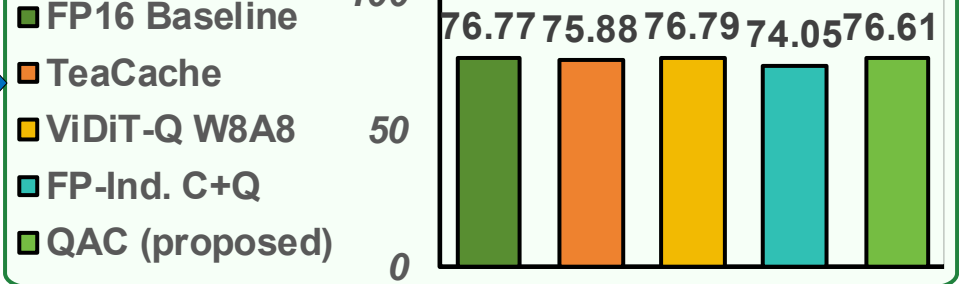


One dot-product result

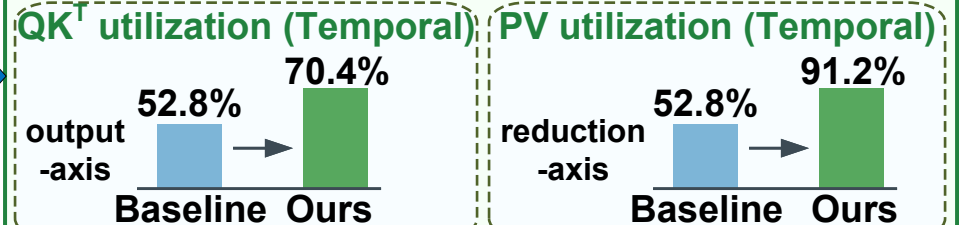
Proposed: boundary-aware segmented reduction



## 1 Algorithm Quality



## 2 Hardware Efficiency



## 3 System-Level Gains

STDiT Step Throughput: 10.96 $\times$  vs A100

Video Throughput: 1.06 $\times$  vs ViDA

Energy Efficiency: 26.59 $\times$  vs A100

Area Efficiency: 2.04 $\times$  vs ViDA